Electrical Engineering Portfolio

Alfie Saji Varghese | Office lighting and electrical layout study

This portfolio presents my work on an office lighting and electrical layout study. I used DIALux to develop and assess a luminaire layout, and Bluebeam to prepare drawing markups and circuit information.

Project scope

The building includes an open office, meeting rooms, private offices, washrooms, a corridor and support rooms. My work covered luminaire research, lighting calculations, socket placement and drawing notation.

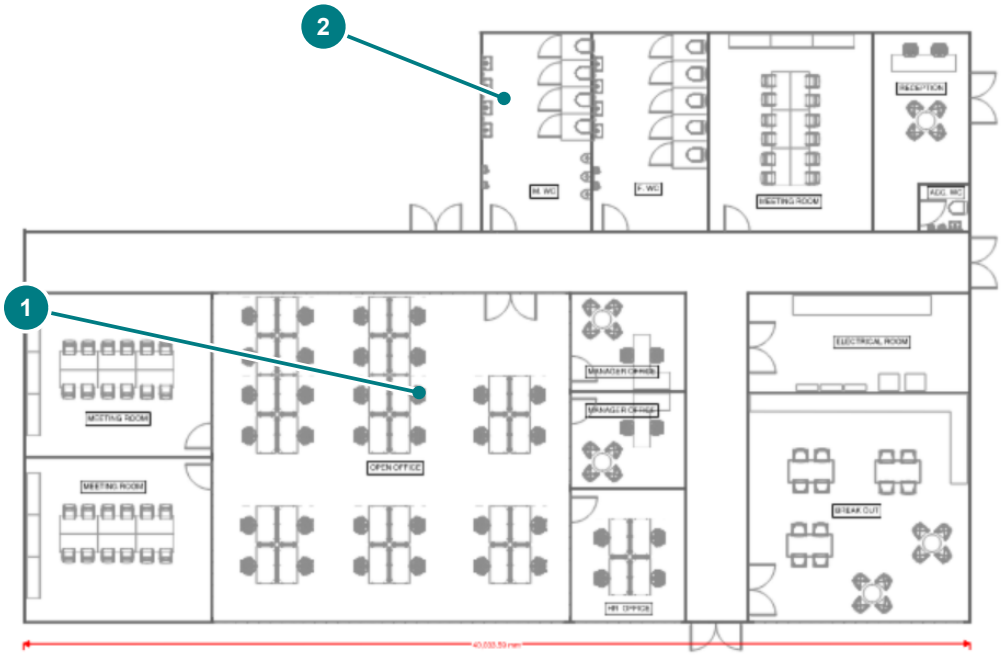


Figure 1 Office floor plan used as the basis for the study.

1 Open office workstations inform the layout. 2 Washrooms need fittings suited to their environment.

My contribution

- Selected luminaires and adjusted their quantity in response to room lighting needs.
- Used calculation results and false colour plots to examine light levels and distribution.
- Prepared lighting and electrical markups, circuit references and room schedules.

Project tools DIALux and Bluebeam
Portfolio focus Lighting design, electrical layouts and technical learning

Design learning exercise. Drawings and software results show the development work recorded in my notes.

Lighting layout

Luminaire selection and positioning

I researched suitable luminaires for the office and adjusted their quantity according to the lighting needs of each room. I considered the ceiling construction and the activities taking place below each fitting.

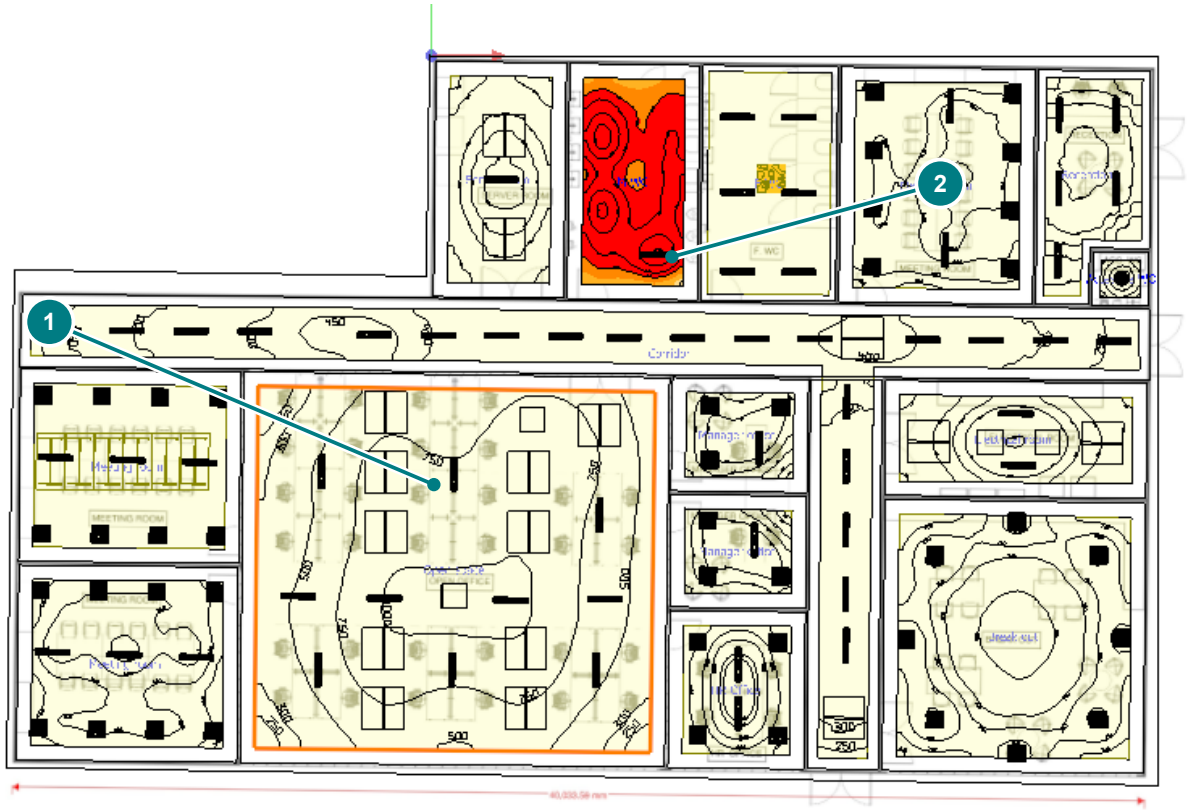


Figure 2 DIALux lighting layout with illuminance contours.

1 Luminaires positioned across the workspace. 2 Contour lines connect equal illuminance values in lux.

Design considerations

Ceiling construction

I learnt to distinguish between suspended and solid ceilings and the different luminaire mounting arrangements they require.

Room conditions

I researched ingress protection for washroom luminaires, considering exposure to water when selecting a suitable fitting.

Task lighting and colour temperature

I considered lighting above work surfaces to improve visibility. I also explored warmer white light for washrooms and a cooler white appearance for working areas.

Lighting calculation review

Calculated illuminance and light distribution

I used DIALux calculations to guide changes to the number of luminaires. False colour plots helped me see how light spread across a room and identify areas with lower illumination.

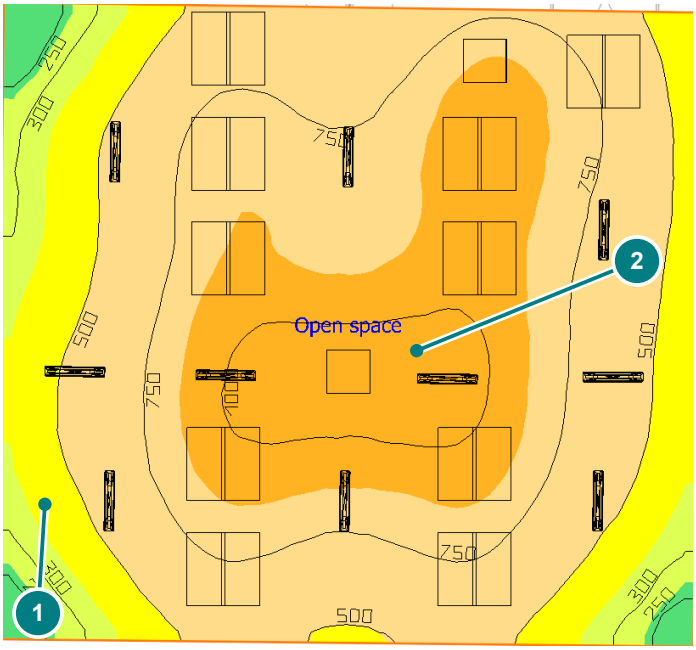


Figure 3 Open office false colour study. Callouts identify the perimeter and central workspace.

Reading the plot

1 The perimeter shows lower light levels than the central area.

2 Higher light levels occur within the workspace.

The labelled contours show illuminance in lux. These are calculated values.

Selected recorded results

Space in the result panel	Illuminance shown
Meeting room entries	502, 532 and 518 lx
Open space	684 lx
HR office	561 lx
Reception	301 lx

Transcribed from the retained DIALux result panel. The three meeting room entries are retained as shown because the room names are truncated in the screenshot.

What I learnt

An average light level does not describe the whole room. I also need to consider the distribution of light across the working area. The retained results document my calculation review; they do not establish a final compliance assessment.

Drawing coordination

Bluebeam markups and circuit identification

I used Bluebeam to mark up lighting and electrical layouts on the office plan. Circuit labels and schedules helped connect the drawing information to the rooms and equipment being served.

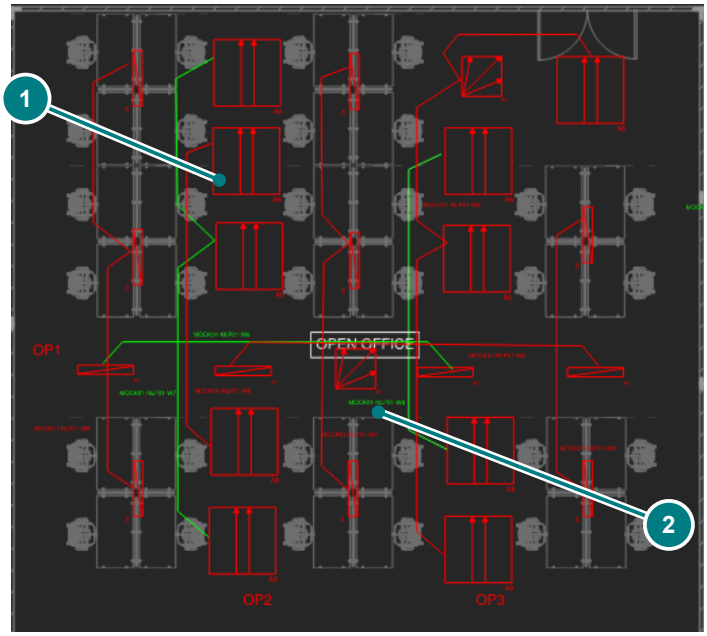


Figure 4 Open office lighting detail.
1 Luminaire positions relative to the desks. 2 Circuit routes and labels within the office.

Reading a circuit reference

MOCK01-NLP01-W5

MOCK01 identifies the project. NLP01 is the normal power board designation used in the study. W5 identifies the circuit way.

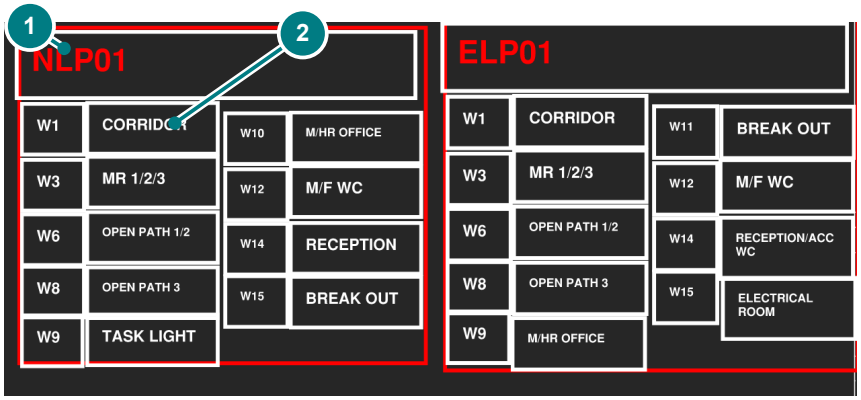


Figure 5 Board schedules.
1 Board identifier. 2 Circuit way and room allocation, linking the schedule to the plan.

I also learnt to read mounting height notes: FFL means finished floor level and TOF means top of fixture. A note of 3 m from FFL to TOF describes the fixture height relative to the finished floor.

Electrical power layout

Equipment connections and socket placement

I placed sockets for office equipment and other appliances, relating their positions to the desks and room functions. I also learnt about cleaner sockets for vacuum cleaners and fused connection units for appliance connections.

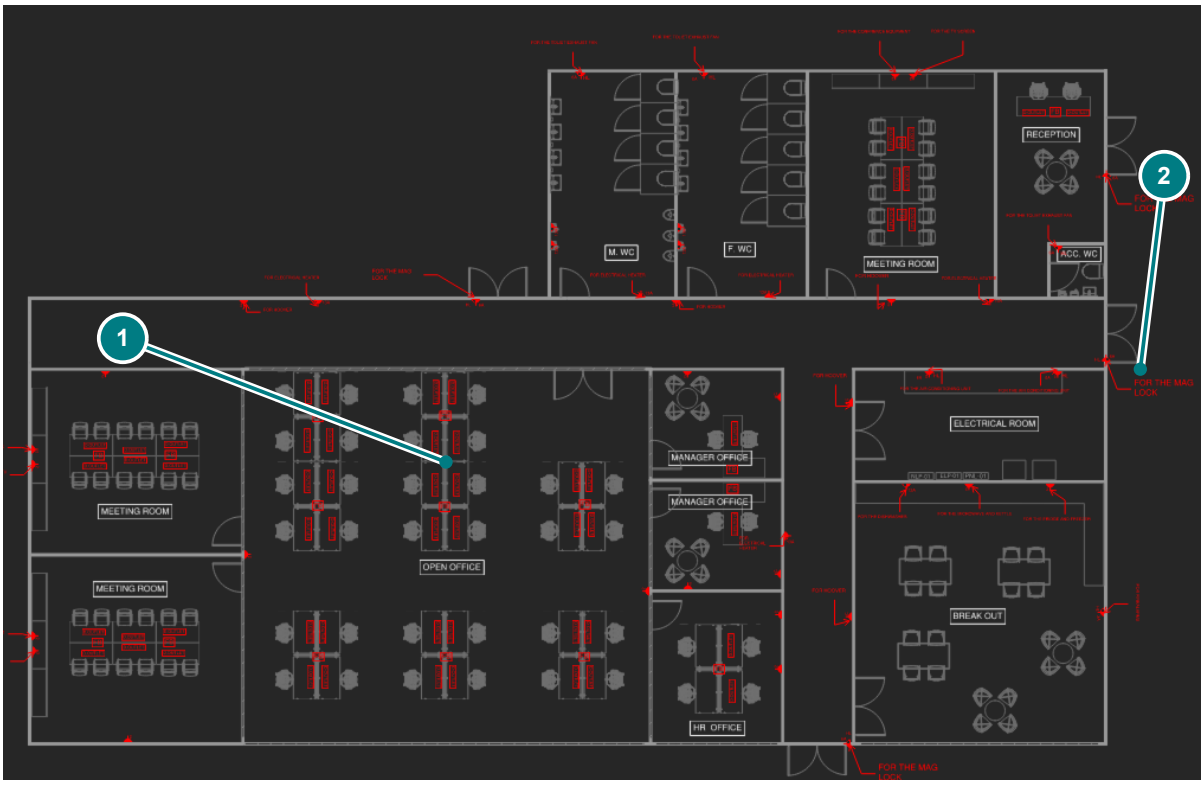


Figure 6 Socket and equipment connection layout.

1 Connection points distributed at workstations. 2 Perimeter connection locations.

Connection choices

Fused connection units
I explored switched and unswitched FCUs. The appliance connection and isolation requirements influence the choice.

Equipment loads
I reviewed indicative appliance power values to understand the range of loads within an office, including kitchen equipment, IT devices and building services.

Distribution learning

I explored busbars as conductors used to distribute high currents. Their resistance and thermal performance depend on the material, cross section and connections. Resistive power loss follows $P = I^2R$.

Technical learning

AC power and maintained lighting performance

Power factor

I learnt the distinction between real power, reactive power and apparent power. For a single phase circuit with sinusoidal voltage and current, using RMS values:

Apparent power $S = VI$ (VA)
Real power $P = VI \cos \varphi$ (W)
Power factor $P / S = \cos \varphi$

The angle φ represents the phase difference between voltage and current. For a given real power and voltage, a lower power factor means a higher current, which affects loading and heating. Apparent power therefore matters when considering the capacity of electrical equipment. [2]

Maintenance factor

I learnt that lighting performance changes over time as light sources age and dirt accumulates. The maintenance factor accounts for these effects in the calculation. Cleaning intervals, ambient conditions and the performance of the light sources influence the factor used. [1]

Project reflection

This study helped me connect software results with drawing decisions. I developed confidence in DIALux calculation review and Bluebeam markups, while improving my understanding of luminaire selection, circuit identification and equipment connections.

The key lesson was to read the calculations alongside the layout: the location of desks, the room use and the spread of light all affect how a design should be assessed.

Technical references

[1] DIALux Knowledge Base. Maintenance factors.
[2] Schneider Electric Electrical Installation Guide. Definition of reactive power.